

LoRa®-based Wireless Software Package

Introduction

- Radio Driver
- RadioEvalApp
RadioEvalApp
- Ping-pong
- LPWA Power Estimator
- LoRaWAN MAC
- LoRaSample
- LoRaFuotaSample
- LoRaSensorSample
LoRaSensorSample RA2E1, RA2L1, RA0E1, RA0E2 LoRaWAN
Sensor Demo
- LoRaWanPrivateLoRaComboSample
- PrivateLoRaSample

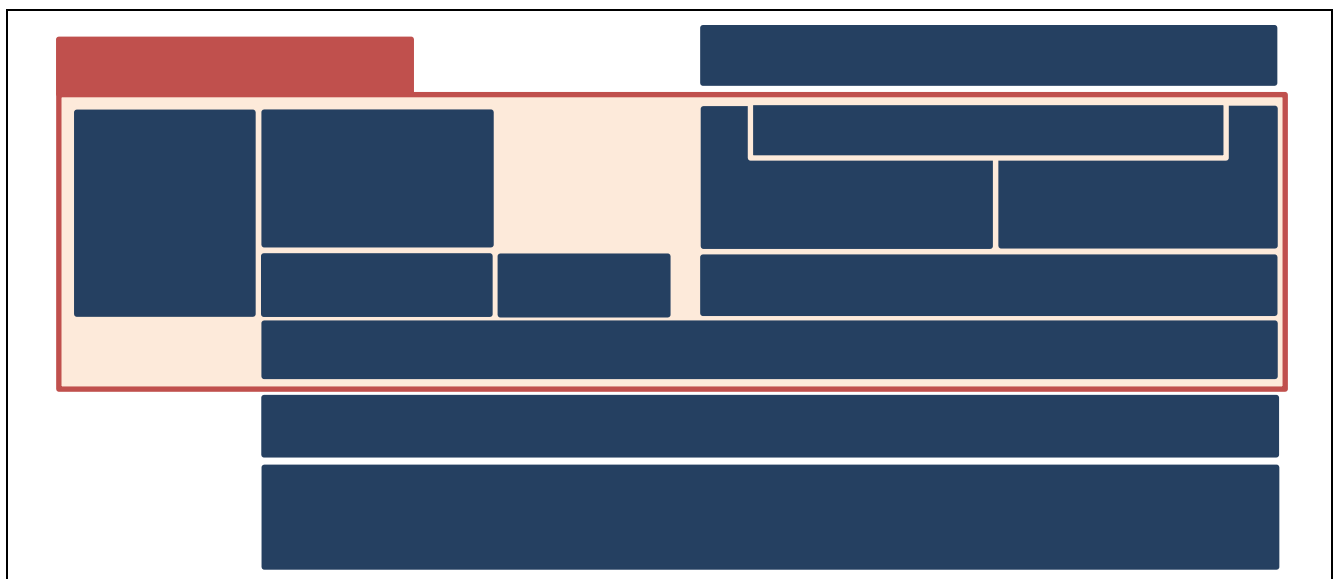


Figure 1. Sample Software and Tools in Current Software Package

Target Device

Contents

1. Hardware Setup

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References:

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Important:

1.1 Hardware Setup for FPB-RA2E1 Fast Prototyping Board

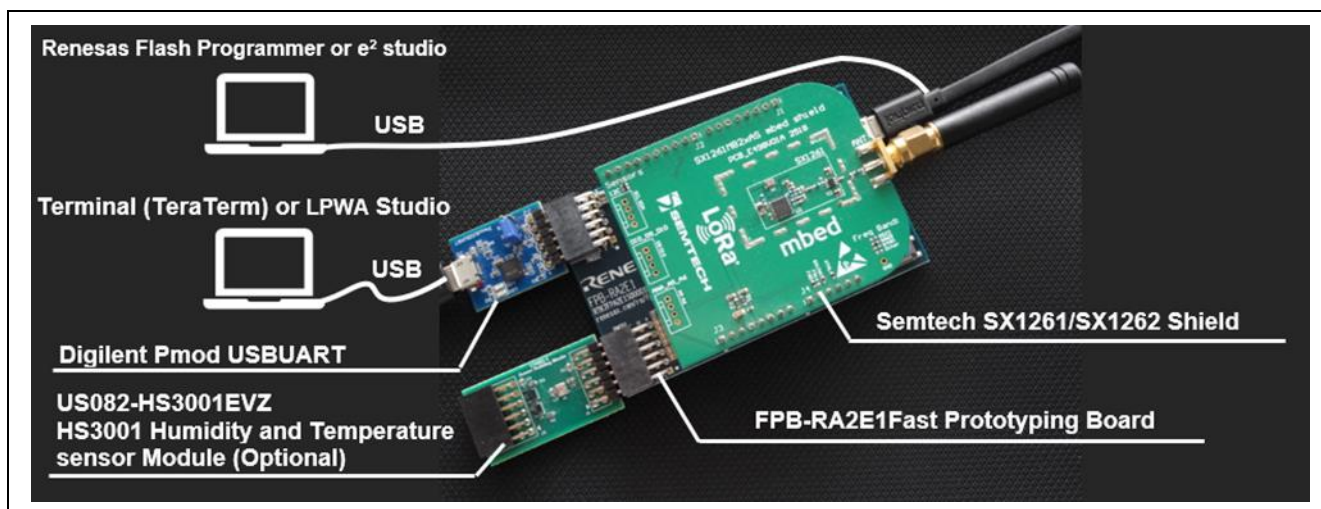


Figure 2. FPB-RA2E1 Hardware Setup Example

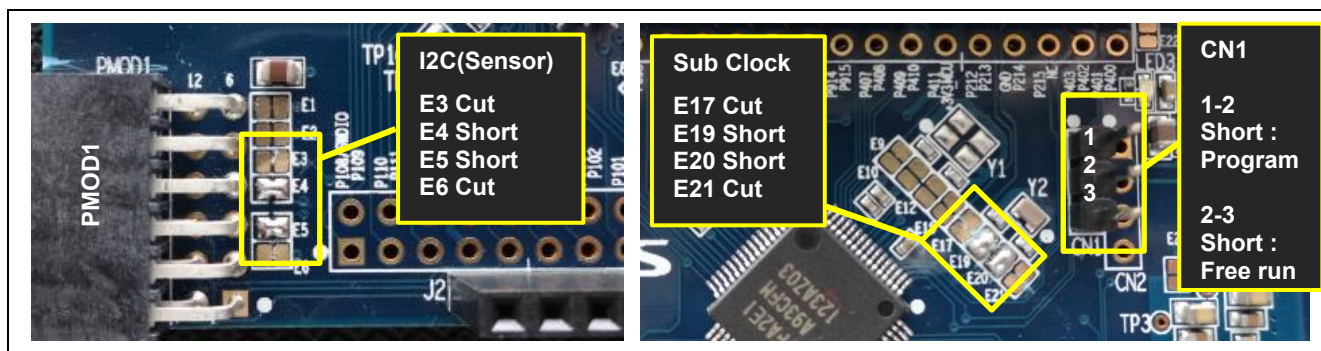


Figure 3. FPB-RA2E1 Configuration Pins

1.2 Hardware Setup for EK-RA2L1 Evaluation Kit for RA2L1 MCU Group

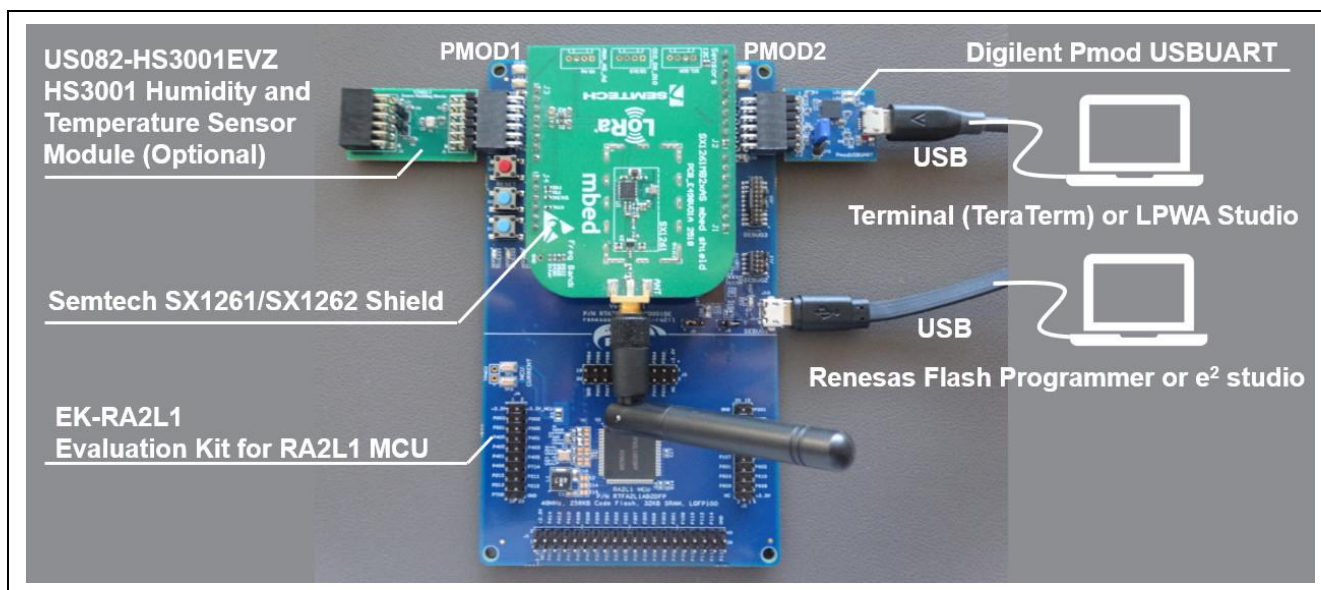


Figure 4. EK-RA2L1 Hardware Setup Example

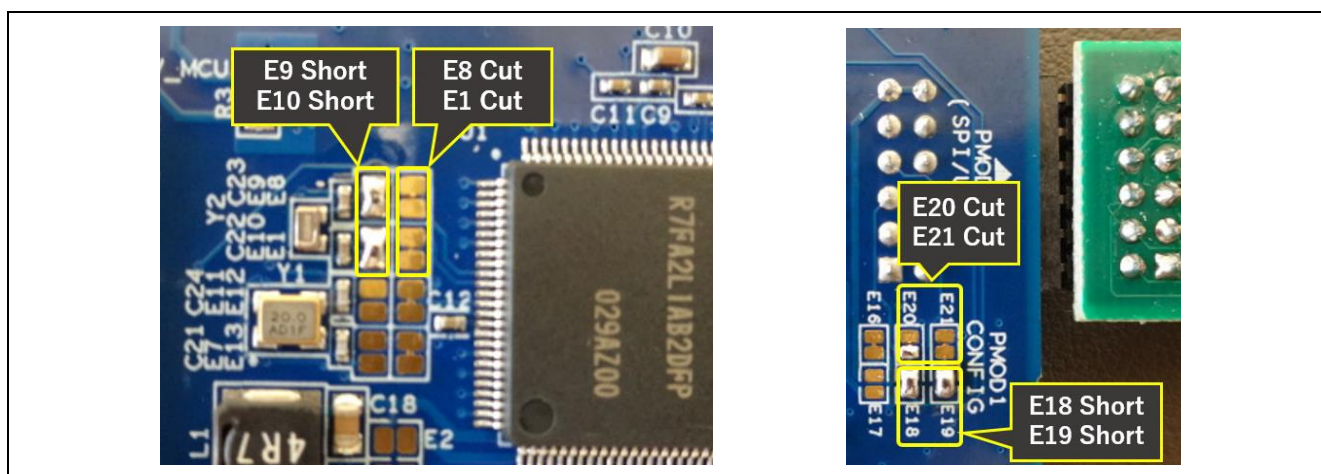


Figure 5. EK-RA2L1 Configuration Pins

1.3 Hardware Setup for FPB-RA0E1 Fast Prototyping Board

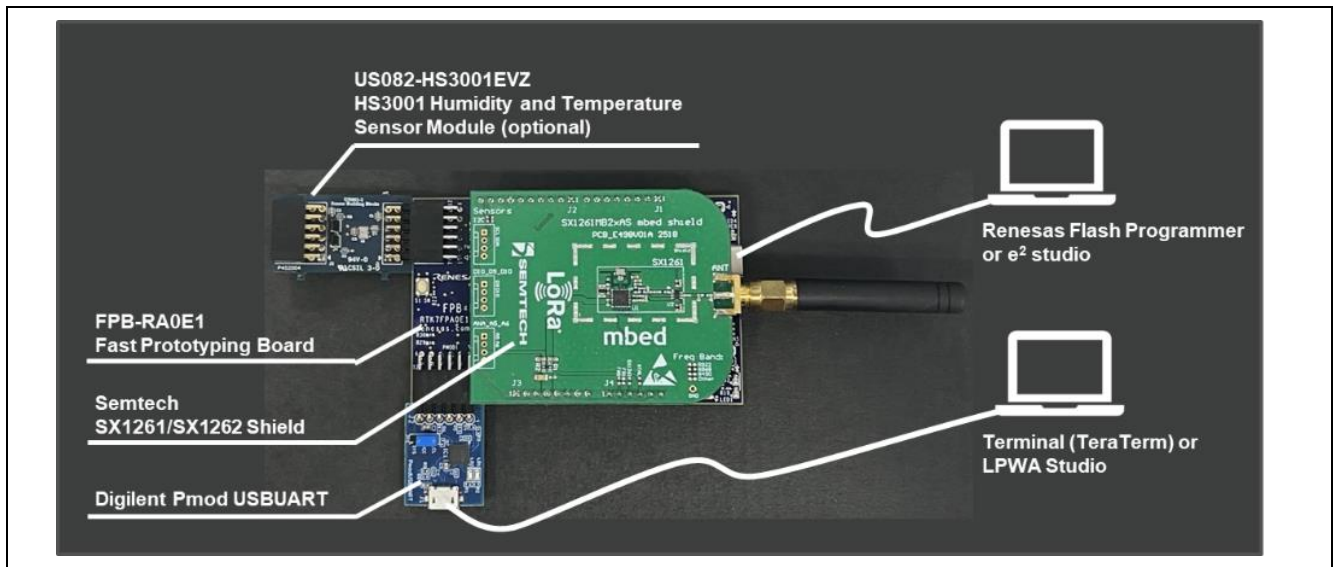


Figure 6. FPB-RA0E1 Hardware Setup Example

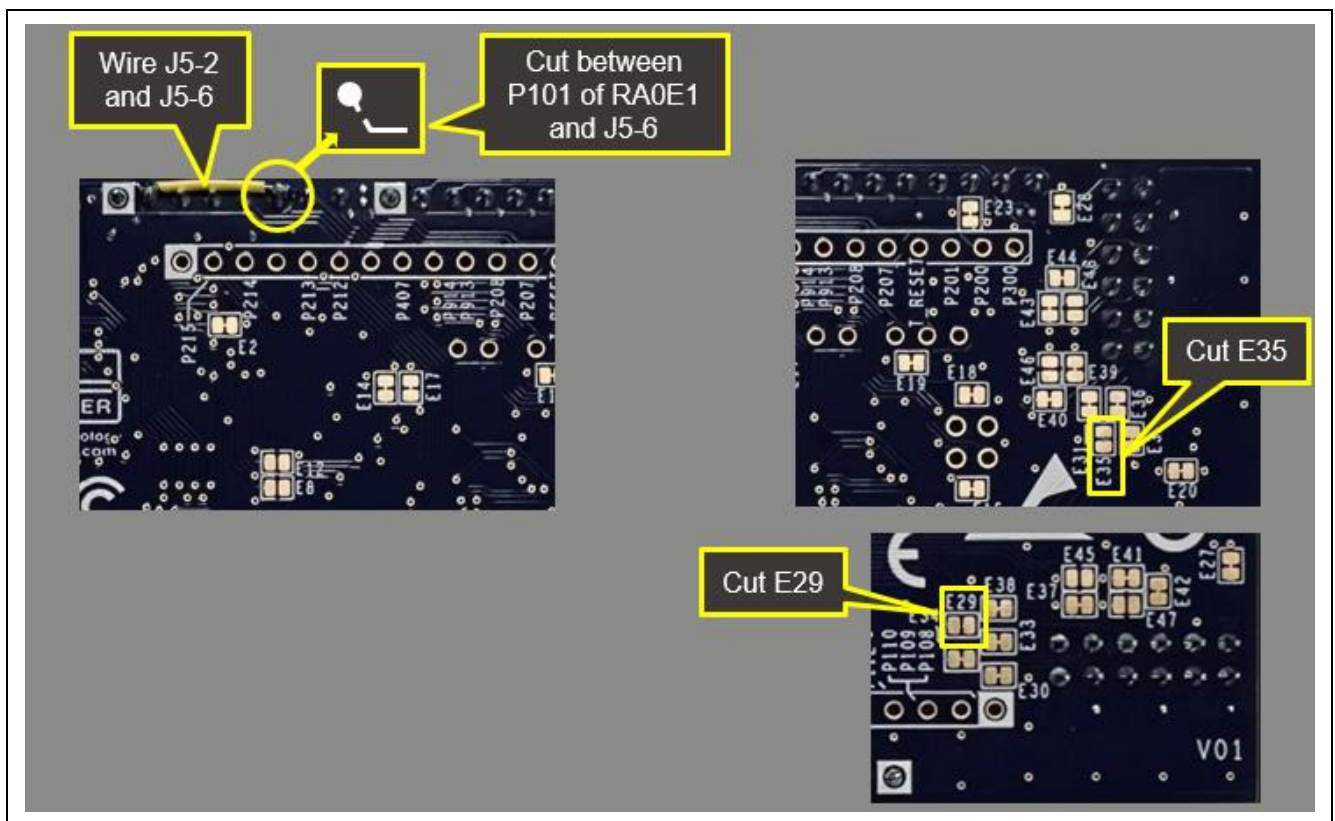


Figure 7. FPB-RA0E1 Configuration Pins

1.4 Hardware Setup for FPB-RA0E2 Fast Prototyping Board

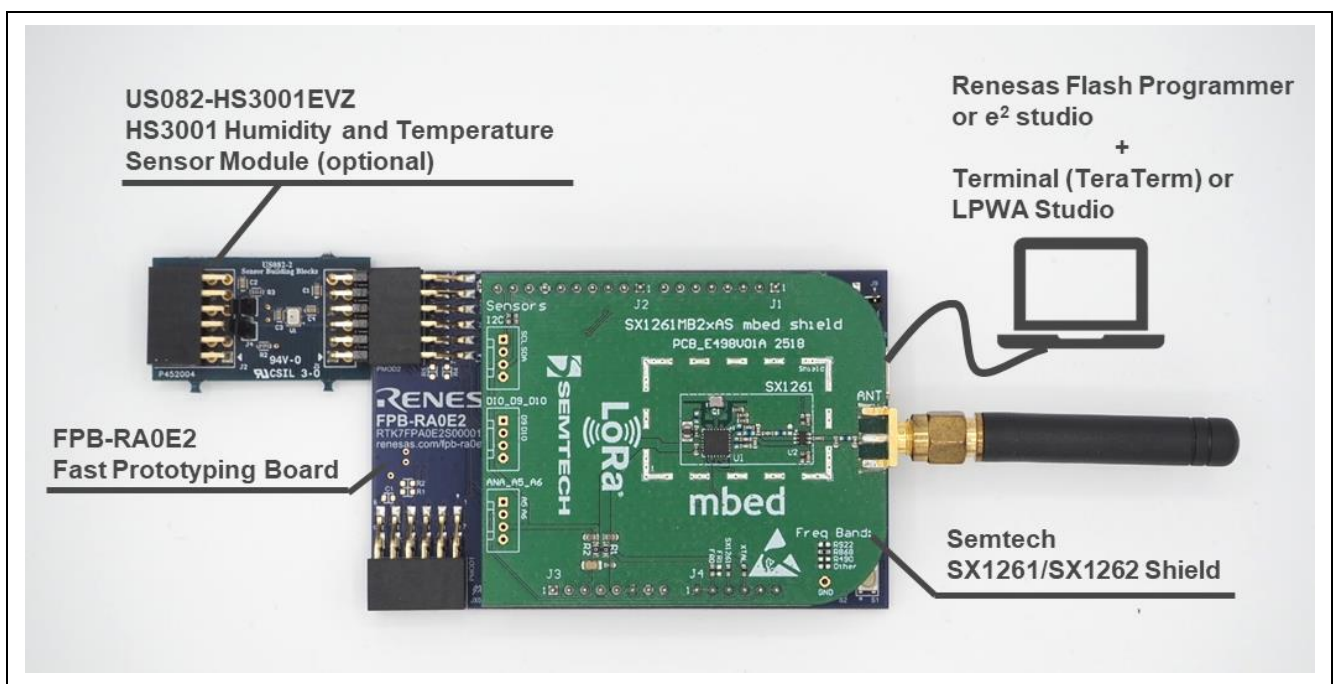


Figure 8. FPB-RA0E2 Hardware Setup Example

2. Sample Application Software Setup

- RadioEvalApp
- ping-pong
- LoRaSample
- LoRaFuotaSample
- PrivateLoRaSample
- LoRaWanPrivateLoRaComboSample

LoRaSensorSample	RA2E1, RA2L1, RA0E1, RA0E2 LoRaWAN Sensor
Demo	
LoRaFuotaSample	
LoRaWanPrivateLoRaComboSample	

2.1 Virtual COM Port Driver Setup

_____	setup executable
CDMxxxx_Setup.zip	

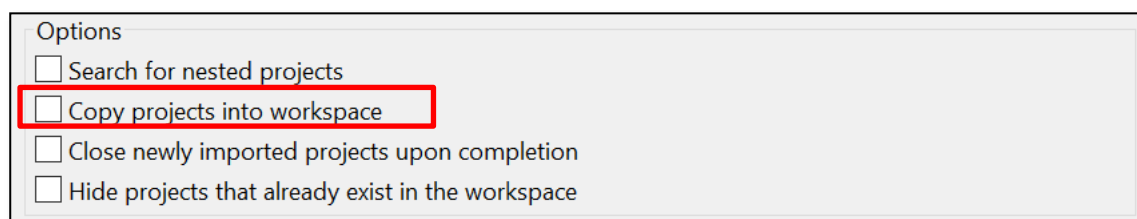
2.2 Software Setup to run Pre-compiled Sample Applications

Renesas Flash Programmer V3.21.00

	(.hex
File	New Project ..
—	RA
—	sample1
—	C:\Temp\rfp
—	E2 emulator Lite (for FPB-RA2E1) J-LINK (for EK-RA2L1, FPB-RA0E1 or FPB-RA0E2)
	RadioEvalApp.hex
	\samples\project\e2studio\{BOARD}\{APPS}\{APPS}.hex
{BOARD} :	ra2e1fpb_sx126x, ra2l1ek_sx126x, ra0e1fpb_sx126x or
	ra0e2fpb_sx126x
{APPS} :	RadioEvalApp, LoRaSample, LoRaFuotaSample, PrivateLoRa,
	LoraWanPrivateLoRaComboSample(, LoRaSensorSample).
Start	File Exit.

2.3 Software Setup for Developing Software Based on the Sample Applications

File	Import	Next.
General	Existing Projects into workspace	Finish
Browse	root directory	
\samples\project\e2studio\{BOARD}\{APPS}\		
{BOARD} : ra2elfpb_sx126x , ra21lek_sx126x, ra0elfpb_sx126x or ra0e2fpb_sx126x		
{APPS} : RadioEvalApp, ping-pong, LoRaSample, LoRaFuotaSample, PrivateLoRa, LoraWanPrivateLoRaComboSample (, LoRaSensorSample)		



configuration.xml	Project Explorer
Generate Project Content	FSP Configuration
	 

3. Utility Tools Setup for Windows® 10/11

- Renesas LPWA Studio (Windows GUI frontend for RadioEvalApp)

\samples\tools\RLPWASudio\setup.exe

!M ! ! !

- Renesas LPWA Power Estimator (Windows Excel application program)

\samples\tools\RLPWApowerEstimator\RLPWApowerEstimator.xlsx

4. Resources Usage

4.1 Memory Resource Usage

Application	FPB-RA2E1				EK-RA2L1			
	Optimization: -O2		Optimization: -Os		Optimization: -O2		Optimization: -Os	
	ROM (KiB)	RAM (KiB)	ROM (KiB)	RAM (KiB)	ROM (KiB)	RAM (KiB)	ROM (KiB)	RAM (KiB)

Application	FPB-RA0E1		FPB-RA0E2			
	Optimization: -Os -flto		Optimization: -O2		Optimization: -Os	
	ROM (KiB)	RAM (KiB)	ROM (KiB)	RAM (KiB)	ROM (KiB)	RAM (KiB)

RP_USE_RADIO_CFG_CHECK

LORAMAC_CLASSB_ENABLED

LORAMAC_CLASSB_ENABLED

LORAMAC_CLASSB_ENABLED

LORAMAC_CLASSB_ENABLED

LORAMAC_CLASSB_ENABLED

4.2 Peripheral Resource Usage

Resources	Function	FPB-RA2E1	EK-RA2L1

Resources	Function	FPB-RA0E1	FPB-RA0E2

Revision History

Rev.	Date	Description Page
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General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

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